ROBOCUPJUNIOR RESCUE (LINE/MAZE) 2026

TEAM DESCRIPTION PAPER

TOKAI Ghost

Abstract

We are TOKAI Ghost, a team from Japan. Our new robot for this competition is a rescue robot designed to consistently achieve high scores in all situations. At Japan Open 2026, we achieved full marks in every round.

This robot is entirely self-built. Unlike our robot from last season, we didn't use any kits such as LEGO for this robot. We also designed all the electronic components.

Our robot has many innovative features. We developed silicone wheels and a twisting mechanism to navigate difficult ramps and bumps. For line following, we use not only sensors but also a camera this year, enabling more stable tracking. In the evacuation zone, the robot can reliably pick up and efficiently carry victims with its multifunctional rescue mechanism.

For victim detection, we trained a YOLOv8-based model on a dataset annotated using Label Studio. Since it was designed for inference on a Raspberry Pi 5, we were able to develop a highly accurate model.

By combining the strengths of a camera and multiple sensors, the robot improves both speed and stability in line following and in rescue tasks.

1. Introduction

a. Team

- **Takumi Takano** – Captain, Hardware Engineer, Software Developer
  Takumi Takano has participated in RoboCup Junior Rescue Line since 2025. He won 1st place at RCJ Japan Open 2025 and 2026, and achieved 5th place at RoboCup 2025 Salvador. He designed all electronic components and most of the robot's structure, developed the line-following and rescue algorithms, managed the development schedule, and coordinated communication among team members.

- **Hongyi Dai** – Software Developer
  Hongyi Dai has participated in RoboCup Junior Rescue Line since 2025. He won 1st place at RCJ Japan Open 2025 and 2026, and achieved 5th place at RoboCup 2025 Salvador. He developed the evacuation point detection algorithm and a camera-assisted line-following system. He was also responsible for setting up the Raspberry Pi 5 environment and configuring the camera system.

- **Shinichi Kado** – Software Developer
  Shinichi Kado joined the team this season. He developed the YOLOv8-based victim detection system and created efficient methods for image collection, annotation, and data augmentation. He also trained and evaluated multiple detection models, contributing to reliable victim detection and the adoption of new AI technologies within the team.

2. Project Planning

a. Overall Project Plan

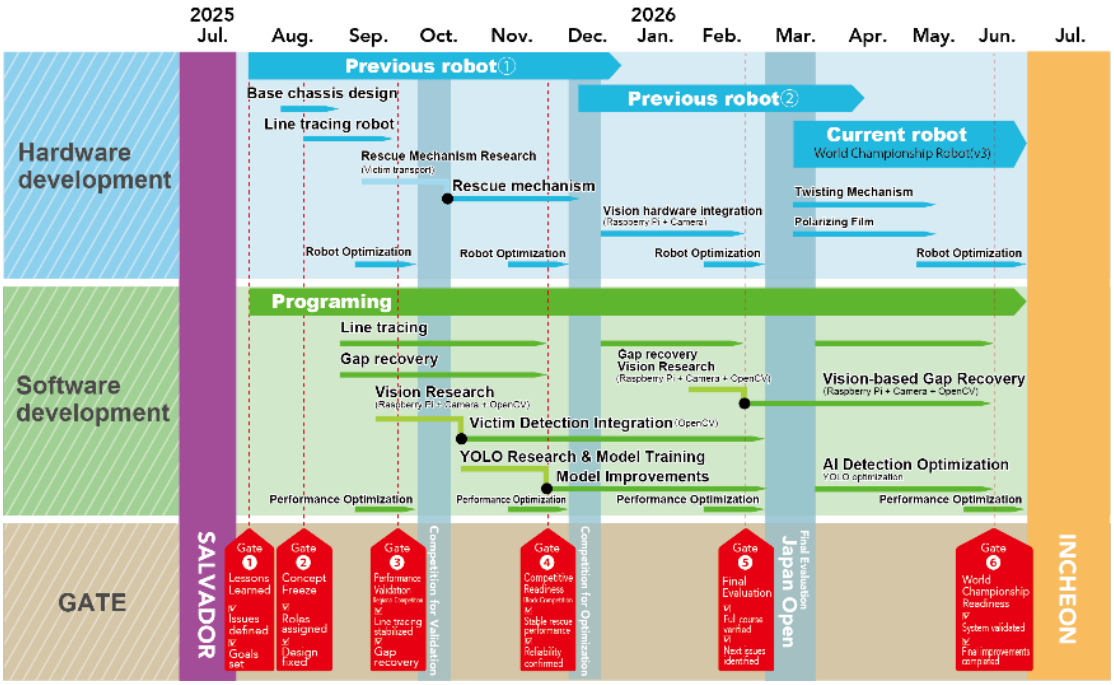
Our goal this year is to achieve a top ranking at the world competition. We think that avoiding LoP under all conditions is more important than speed. To prevent LoP, both high-performance hardware and stable software are essential. Specifically, this includes a multifunctional rescue mechanism, a powerful drive system, optimal sensor placement, and reliable, bug-free software. Our robot had many LoPs at RoboCup 2025. However, at Japan Open 2026, it achieved full marks in every round without any LoPs. This shows that the robot is stable and reliable. Since the field at the world competition is more complex, we think it is necessary to further improve the robot's drive system performance and speed. To achieve this, we planned our development schedule with sufficient time for testing and improvement.

- **Milestones and Project Timeline:**
  - **Powerful Driving System:**
    - **Description:** Created a drive system using powerful servos and metal components to navigate ramps and bumps easily.
    - **Team Member:** Takano
    - **Deadline:** October 12
  - **Evacuation Point Detection:**
    - **Description:** Implemented an algorithm for detecting evacuation points that accounts
    - for obstacles.
    - **Team Member:** Dai
    - **Deadline:** November 29
  - **Multifunctional Rescue Mechanism:**
    - **Description:** Designed a rescue mechanism equipped with various sensors to handle a wide range of situations.
    - **Team Member:** Takano
    - **Deadline:** January 19
  - **Victim Detection with Machine Learning:**
    - **Description:** Developed YOLO-based models for accurate detection.
    - **Team Member:** Kado
    - **Deadline:** February 22
  - **Camera-Assisted Line Following System:**
    - **Description:** Implemented a camera-based assist system to compensate for the limitations of sensor-based line following.
    - **Team Member:** Dai
    - **Deadline:** April 25
  - **Twisting Mechanism:**
    - **Description:** Created a twisting mechanism to improve the stability of line following.
    - **Team Member:** Takano
    - **Deadline:** May 7
  - **LED Glare Reduction with Polarizing Film:**
    - **Description:** Tested the camera with a polarizing filter and compared performance.
    - **Team Member:** Dai, Kado
    - **Deadline:** May 10

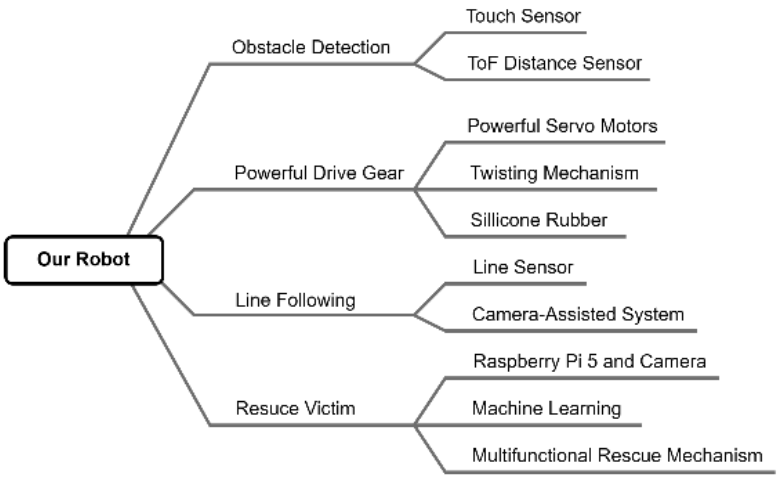
- **Schedule/Plan Agreement Process:**
  All team members discussed the requirements for the robot, and our captain coordinated the team's ideas and decided on the development schedule. We also frequently shared progress updates via Discord to ensure efficient development.

b. Integration Plan

When adding new components, we first tested them in a separate test environment, then integrated them into the robot, and finally verified their performance again.
Throughout the season, we ensured hardware compatibility between the current model and the previous model. Most parts of the current robot are identical to the previous version. Therefore, the same control program could be used on both robots, allowing software development and testing to continue while the new hardware was still under development.
As a result, hardware and software could be developed in parallel.



We designed large wheels to easily climb ramps and overcome bumps. At the same time, we minimized the overall size of the robot to gain an advantage within the narrow passages defined by the rules. We adopted a Raspberry Pi 5 and a camera for accurate victim detection. We used separate microcontrollers for different functions to ensure stable sensor processing. All components communicate via UART for reliable system integration.

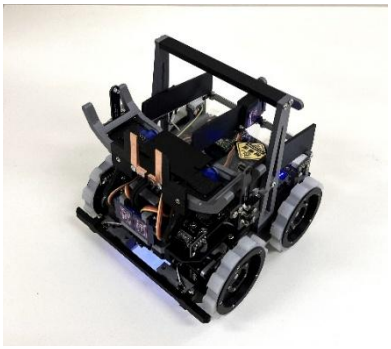
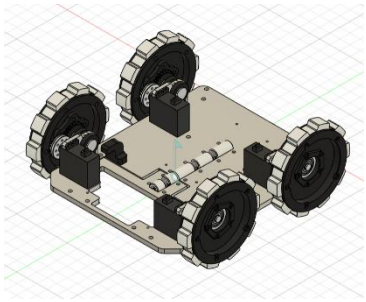
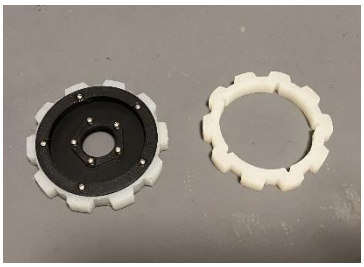


3. Hardware

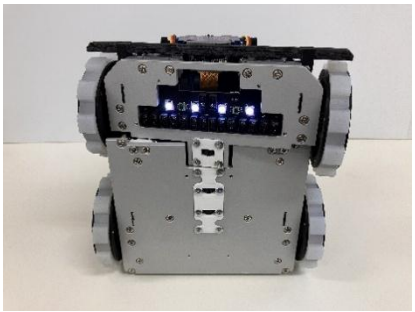
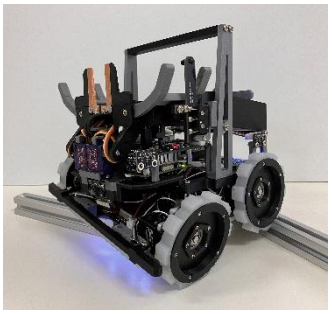
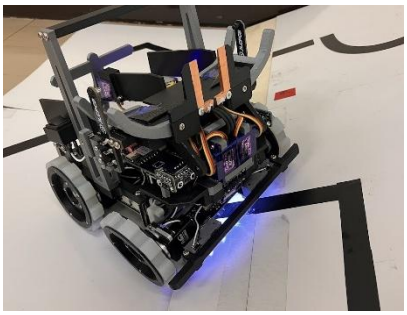
a. Mechanical Design and Manufacturing

- **Main Structure:**
  This is an image of our robot.
  To improve space efficiency, we decided from the beginning to design the entire robot structure ourselves and manufacture most of it using 3D printing. We designed the robot using Autodesk Fusion. The robot measures 163 mm × 174 mm.
  An aluminum chassis is used at the bottom of the robot to ensure overall durability and reliability. To lower the robot's center of gravity, we used more aluminum components in the lower half and more 3D-printed components in the upper half.

- **Actuators and Power Train:**
  We use four STS3032 serial servo motors. These motors are compact yet powerful, allowing the robot to effectively overcome bumps.
  The wheels have a large diameter and a narrow width. The outer surface of the wheels is made by molding and curing silicone in a custom shape. This shape provides high grip on bumps, including those on ramps.

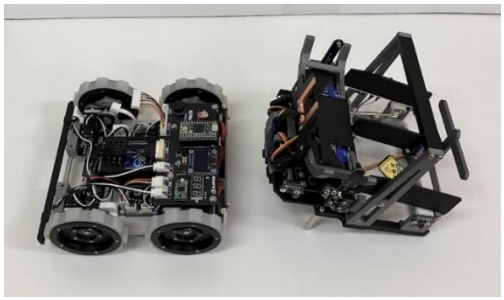
  In addition, we developed a twisting mechanism. The front and rear halves of the robot are connected using linear shafts, allowing a relative twisting motion between them. By enabling this motion, all wheels maintain contact with the floor even on uneven terrain. This improves the robot's driving performance on bumps.

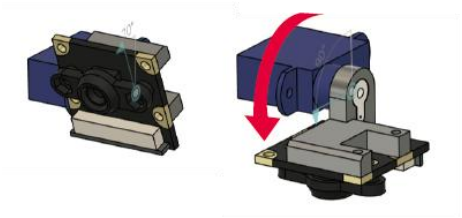




  A key point is that even at the transition between flat tiles and ramps, the distance between the line sensors and the floor remains very stable. This drive system is innovative and enables smooth and stable sensor-based line following.

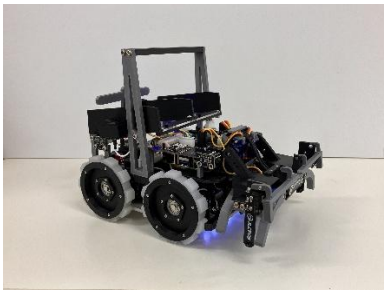
- **Subassemblies/modules:**
  The robot is divided into a lower line-following section and an upper rescue section.
  The lower section consists of line sensors, the drive wheels, a bumper for obstacle detection, and the battery. It is responsible for stable navigation and collision detection. The upper section contains the rescue mechanism, a camera, and distance sensors. It handles victim detection and rescue operations.

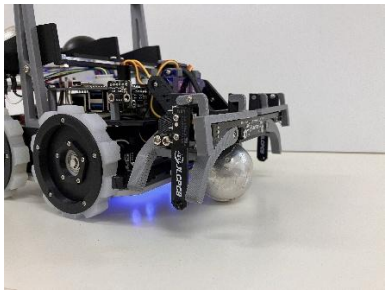
The camera is mounted on a servo mechanism in the upper section and is mainly used for victim detection during rescue tasks. It can also be rotated downward and temporarily used to assist line following in gap situations.
The upper and lower sections are designed to be easily detachable using four screws for maintenance and assembly.

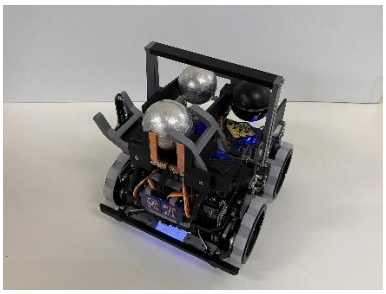




- **Rescue mechanism:**
  We aimed to design a mechanism that can reliably classify all victims and store them simultaneously. This enables faster collection and significantly reduces travel time.
  In addition, we designed a rescue arm that reliably retrieves detected victims. The arm is equipped with multiple sensors that detect contact with obstacles and walls, confirm whether a victim is inside the gripper, and determine victim conductivity. These functions allow the robot to accurately assess the surrounding environment, the timing of victim collection, and the type of victim.
  After lifting a victim, the gripper can open on only one side, allowing classification into different storage locations within the robot. In addition, a tilting gate mechanism enables selective release of each storage section.
  This rescue system integrates multiple functions into a compact robot design. Its functionality and reliability also help reduce software complexity.

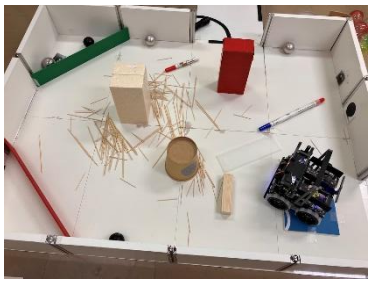






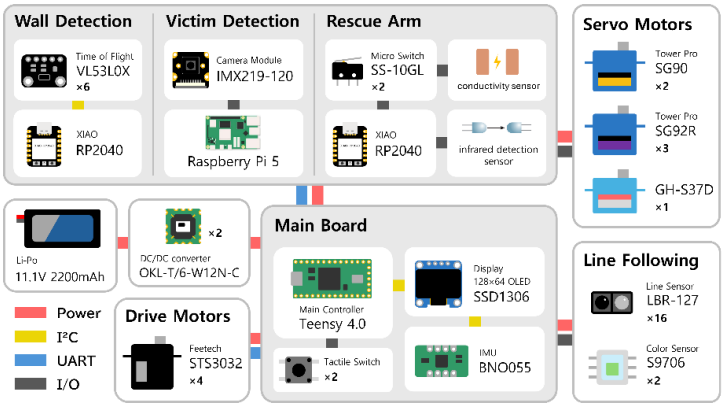
- **Reliability Tests and quality assurance:**
  Threaded metal inserts were used at component connection points, and insert nuts were used in 3D-printed parts. These fastening methods provide strong and reliable mechanical connections.
  For the silicone molding process used for the wheel outer rings, the silicone was degassed in a vacuum chamber before casting to remove air bubbles. This process ensured high-quality silicone parts with minimal internal voids. In addition, several molded parts were intentionally destroyed and inspected to verify their internal quality.

  We conducted extensive rescue tests on the field shown below. The field contained numerous obstacles and bumps, including some conditions beyond those permitted by the official rules. Through repeated testing, we identified and resolved issues in the rescue mechanism and improved the overall durability and reliability of the robot.



b. Electronic Design and Manufacturing

- **Detailed Overview:**
  This diagram shows the electronic components used in our robot.



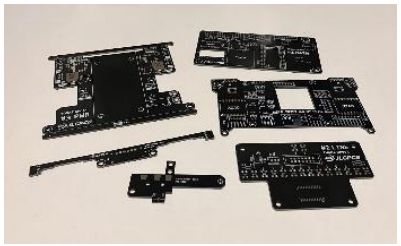
We use sixteen line sensors for line following, and all of them are read as analog values. For color detection, we use digital color sensors, which prevent communication freezes and provide reliable operation. Six VL53L0X distance sensors are mounted on the upper section of the robot for obstacle and wall detection. Last year, we experienced issues with sensor drift. To improve reliability, we adopted the BNO055 gyroscope this year. It is used for ramp detection and accurate movement control. The rescue arm is equipped with an infrared detection sensor, a conductivity sensor, and touch sensors. These sensors are used to detect whether a victim has been captured, identify victim types, and detect contact with surrounding objects. They contribute to reliable rescue operations in challenging field conditions.

We use a Teensy 4.0 as the main controller. Its high processing speed and large number of UART interfaces make it well suited for integrating the entire robot system. We designed our own expansion board and utilize nearly all available I/O pins. We also use a Raspberry Pi 5 for image processing and AI inference. Only the camera is connected directly to the Raspberry Pi 5, and the processed data is sent to the main controller. By combining the Teensy 4.0 and Raspberry Pi 5, the robot can achieve both high-speed control and accurate vision-based recognition. In addition, we use two XIAO RP2040 boards as sub-controllers. These boards handle dedicated sensor and actuator tasks in parallel with the main controller.

The robot is powered by an 11.1 V LiPo battery. To provide sufficient power for the Raspberry Pi 5 and the servo motors, we use two OKL-T/6-W12N-C converters to generate two independent 5 V power lines. This power system allows the Raspberry Pi 5 to operate stably without unexpected shutdowns.

- **Sub-module Design and Reliability Tests:**
  The Raspberry Pi 5, Teensy 4.0, and XIAO RP2040 boards communicate via UART. This architecture separates vision processing, high-speed control, and peripheral management across dedicated processors, enabling parallel processing and improving overall system reliability.

  All PCBs were designed using KiCad. Connectors are used throughout the robot, allowing components to be easily removed for maintenance and replacement. In addition, three PCBs are mounted directly on the rescue arm, with sensors attached directly to the boards. As a result, wiring inside the arm was eliminated. This unusual design allowed us to create a compact and reliable rescue arm assembly.





During rescue operations, we encountered an issue where the Teensy 4.0 would unexpectedly reset. We first suspected accidental activation of the LoP switch, but testing ruled out this possibility. Next, we investigated the possibility of voltage drops. Adding a 470 µF capacitor near the Teensy 4.0 had no noticeable effect. However, adding a 1000 µF capacitor together with a Schottky diode significantly improved the situation.

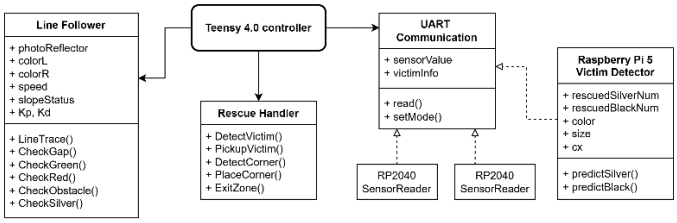
To verify the effectiveness of this solution, we conducted reliability tests under the following conditions. The results are shown below. Adding only a capacitor was not sufficient to eliminate resets. However, the combination of a 1000 µF capacitor and a Schottky diode completely prevented unexpected resets during testing. This solution was therefore adopted in the final robot.

Configuration	Test Runs	Resets
No Countermeasure	20	4
470 µF Capacitor	20	4
1000 µF Capacitor	20	2
1000 µF Capacitor + Schottky Diode	20	0

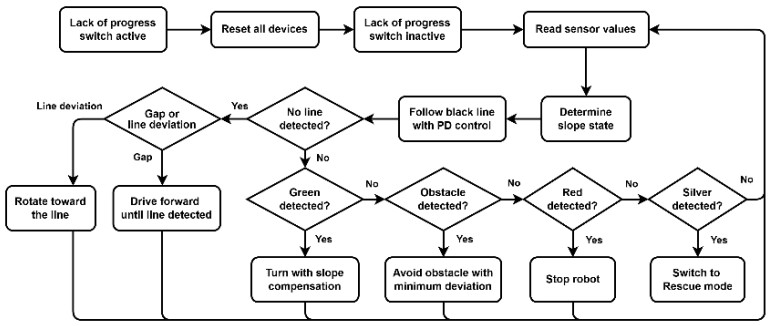
4. Software

a. General software architecture

The programs for the Teensy and RP2040 were developed using the Arduino framework. The Raspberry Pi software was written in MicroPython and utilizes libraries such as OpenCV and Ultralytics. Sensor data and other information are aggregated from the sub-controllers to the main controller, enabling high-speed processing. This processing headroom allows the robot to handle a wide range of situations, achieving stable line following without LoP and reliable victim rescue.

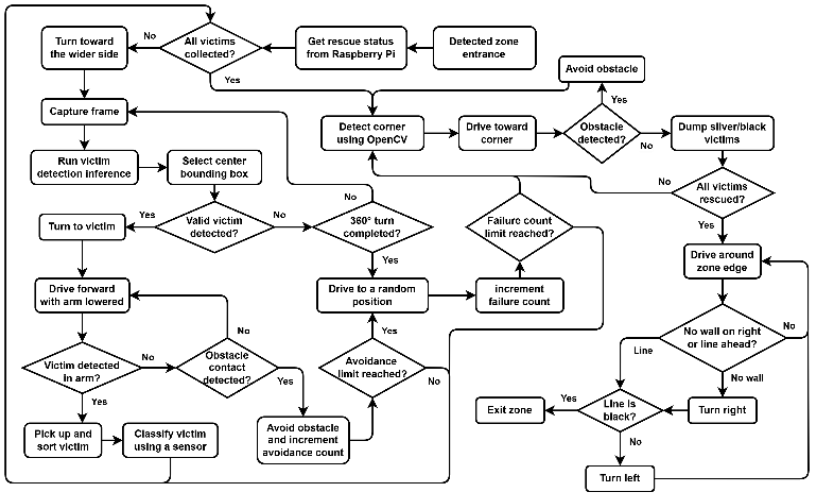


- **Line Following Algorithm:**
  The robot calculates the line position from line sensor data and follows the line using PD control. When it detects an intersection with the color sensor, it determines the appropriate turn based on the elapsed time since the last black line was detected. On ramps, it applies slope-specific compensation to optimize line-following performance for each incline. When no line is detected, it first moves backward and determines whether the situation is a gap or a line following error, significantly reducing the risk of LoP. When encountering an obstacle, it uses distance and touch sensors to navigate around it while maintaining the shortest possible clearance.

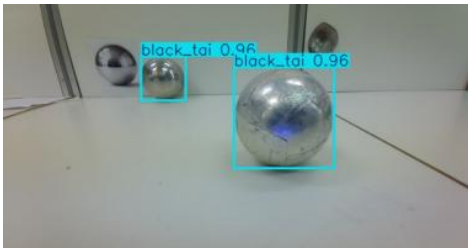


- **Rescue Algorithm:**
  An overview of the evacuation zone process is shown in the figure below. Throughout the rescue operation, the robot utilizes its multifunctional rescue mechanism and distance sensors to handle challenging environments with many obstacles. Because the robot collects victims regardless of

their type, it can perform rescue operations efficiently without prioritizing specific victims. The number of rescued victims is stored on the Raspberry Pi throughout a run. After a LoP, the robot refers to this information, allowing it to perform optimally even on subsequent attempts. The algorithm is designed to handle various situations, such as failing to collect all victims or being unable to locate a corner.

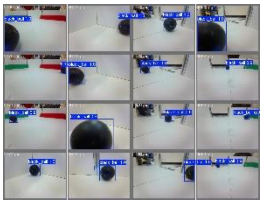


- **Victim Detection:**



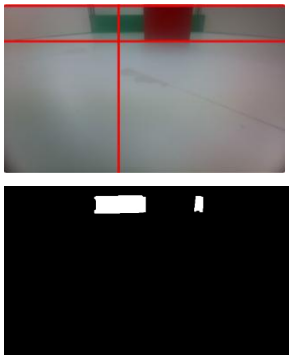
We previously used an OpenCV-based algorithm for victim detection. However, distinguishing victims from the background was difficult. Therefore, we adopted YOLOv8, a deep learning-based object detection model. YOLOv8 detects both the position and type of victims (black balls and silver balls) from images captured by a camera mounted on the robot. We used transfer learning based on the pretrained YOLOv8n model to efficiently train the detector.

  - **Dataset and Training:**
    We collected and processed our own dataset of approximately 3,000 images and manually annotated all victims. To improve robustness and reduce false positives, the dataset included images containing dummy balls, unlabeled images, and Blender-generated images. Data augmentation was also applied to improve performance under different viewpoints and environmental conditions.
    In preparation for the new competition rules, we additionally created and evaluated datasets containing flashlight interference. This allowed us to investigate how different training data affected detection performance under challenging lighting conditions.
  - **Model Evaluation and Selection:**
    To determine the most suitable model for practical use, we trained and evaluated five different detection models. Each model was compared in terms of accuracy (precision, recall, and mAP50), false positive rate, and processing speed (FPS). The evaluation results are shown below. Based on these results, we selected the light-source-augmented model for deployment. This model achieved the best balance between detection accuracy, robustness against flashlight interference, and real-time performance.



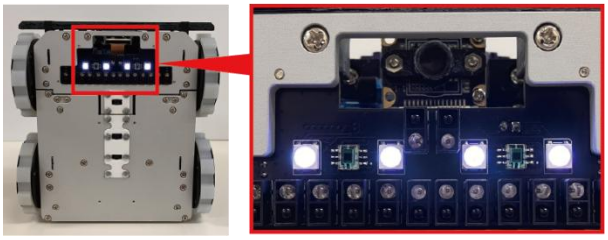
	Precision	Recall	mAP50	FP Rate	FPS
Baseline (Individual Detection)	0.99771	1.00000	0.99442	0.02133	12
Baseline (Simultaneous Detection)	0.98200	0.96730	0.96491	0.10192	8
Blender-Augmented Model	0.99949	0.98629	0.99491	0.02513	8
Light-Source-Augmented Model	0.99453	0.98288	0.99465	0.03421	8

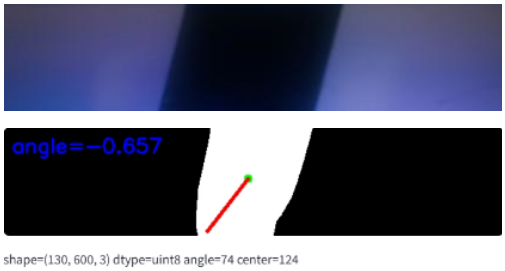
- **Evacuation Point Detection:**
  Because the color and shape of the evacuation point are predefined, it can be detected using simple image processing without AI. First, a mask image is generated by extracting red and green regions from the camera image. Next, connected-component labeling is performed to remove noise. Finally, the correct evacuation point is identified by examining the aspect ratio of the largest remaining label. However, this method may fail when obstacles are present. To address this issue, we consider the two largest labels instead of only the largest one.



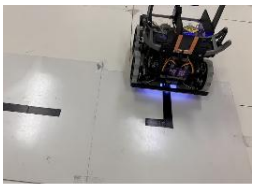
b. Innovative solutions

- **Camera-Assisted Line Following System:**
  We previously used only line sensors for line following. This method has several advantages, such as easy adjustment and low noise. However, in gap tiles, there was a problem: when the straight line before the gap was short, the robot could not align itself with the direction of the line. Therefore, we decided to also use the camera for gap control.





When the robot detects a gap, a servo motor rotates the camera downward. The camera then calculates the angle of the line using PCACompute, a function in OpenCV. By using the calculated line angle for gap control, we found that the robot could avoid losing the line even in difficult gaps. We repeatedly tested the system in difficult gap conditions to evaluate its stability and reliability. The experimental results are shown below.


Hard Gap

Difficulty	With camera	Without camera
Normal	50/50 (100%)	14/50 (28%)
Hard	50/50 (100%)	0/50 (0%)

This approach combines the stability of sensor-based line following with the directional analysis provided by camera-based processing. We believe this system improves the robot's reliability in challenging gap situations.

5. Performance evaluation

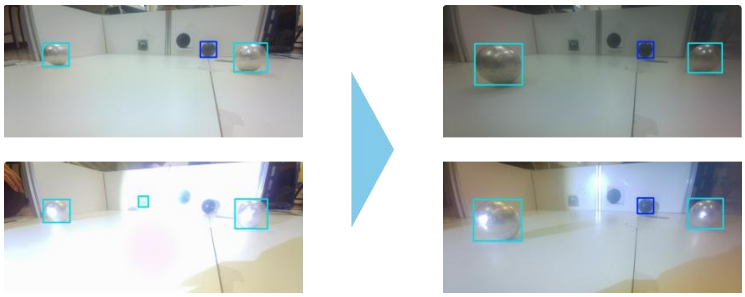
- **High-Speed Line Following Test:**
  We conducted a high-speed line-following test by driving the motors faster than the robot's normal operating speed. The purpose of this test was to evaluate whether the control system remained reliable under demanding conditions.

  - **Slow Control Cycle:**
    During testing, the robot occasionally failed to detect the line, resulting in line-following errors. Detailed investigation revealed that the control cycle was being delayed by the color

sensor readout process. To solve this issue, the color sensor communication was redesigned using timer-interrupt-based asynchronous processing. As a result, the control cycle time was reduced to one-third of its previous value, significantly improving line-following stability.

- **Rescue Test Under New Competition Rules:**
  We conducted rescue tests in an evacuation zone containing flashlight interference and dummy victims. The purpose of these tests was to evaluate the reliability of our solutions for the new competition rules under realistic conditions.

  - **Insufficient Robustness Against Flashlight Interference:**
    Although the victim detection model was trained using images containing flashlight interference, false detections still occurred under certain light intensities and angles.
    To address this issue, we installed a polarizing film in front of the camera lens. The performance improvement obtained using the same detection model is shown below. This modification significantly reduced strong reflections from the floor and improved detection reliability under challenging lighting conditions.



  - **Electrostatic Discharge from Victims:**
    Dummy victims were distinguished from real victims using a conductivity sensor because their appearance and reflectivity were nearly identical. During testing, however, electrostatic discharge from charged victims occasionally caused the microcontroller to reset.
    To solve this issue, we added a comparator circuit and TVS diodes to protect the detection system. This modification eliminated unexpected resets and enabled reliable victim identification.

6. Conclusion

Our new robot features a fully custom-designed compact chassis, a powerful drive system, a multifunctional rescue mechanism, and a line-following system that integrates sensor and camera data. We are confident that these technologies will enable the robot to achieve stable and reliable performance even in challenging competition fields.